Deep Learning-Based Arrhythmia Classification of Individual ECG Heartbeats

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LEEN364 Deep Learning and Classification Techniques · Istanbul Arel University



Why classify single heartbeats?

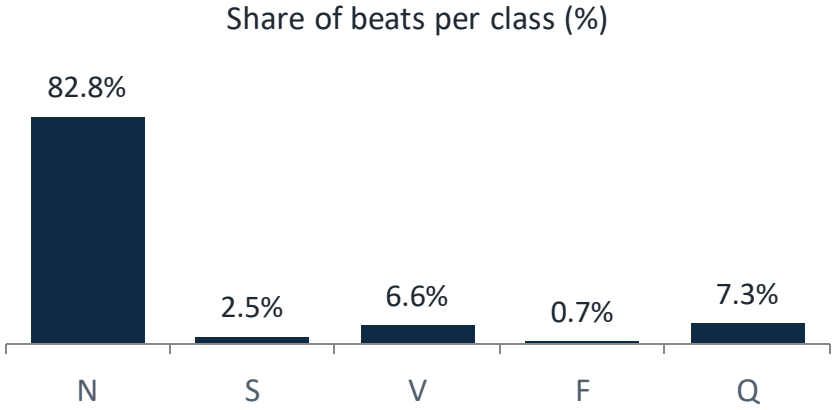
~100,000

heartbeats in one day of a Holter ECG recording

1. **Manual review is slow**
   Rare abnormal beats are easy to overlook.
2. **Goal**
   Classify every beat automatically into one of 5 AAMI classes.
3. **Focus**
   Detect the rare abnormal classes S, V and F, not only normal beats.

N Normal	S Supra-ventricular	V Ventricular	F Fusion	Q Unknown / paced

Dataset: MIT-BIH heartbeats



109,446

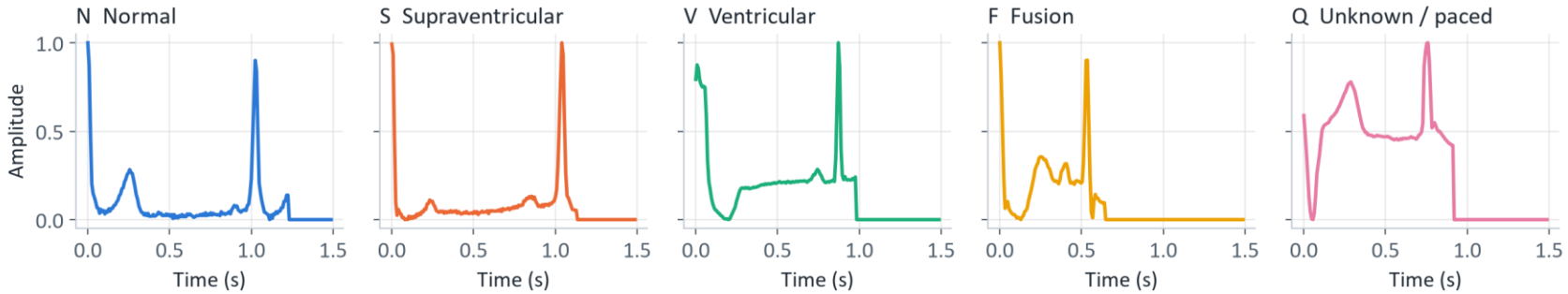
segmented heartbeats

187

samples per beat (1.5 s at 125 Hz)

70k / 17.5k / 21.9k

train / validation / test



Method: from baselines to the final model

1

Split

Train, validation and an untouched test set

2

Baselines

Logistic regression and random forest

3

Neural networks

MLP, simple CNN, residual CNN, LSTM

4

Imbalance study

Class weights, oversampling, augmentation

5

Final evaluation

3 seeds, test set used once

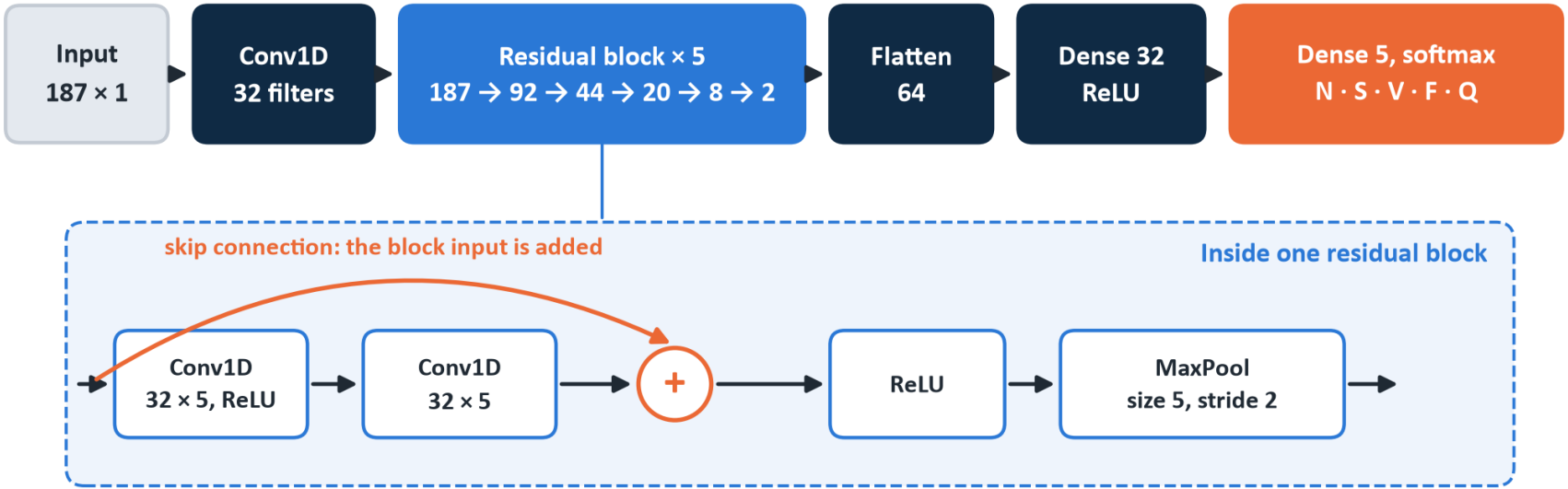
Main metric: macro F1-score

The mean of the F1-scores of the 5 classes, so every class counts equally.

No test-set leakage

All choices on the validation set; oversampling and augmentation on training data only.

Main model: residual 1D CNN



53,957 parameters · Adam, learning rate 0.001 · batch 128 · early stopping on validation macro F1

Handling the class imbalance

B **Class weights**

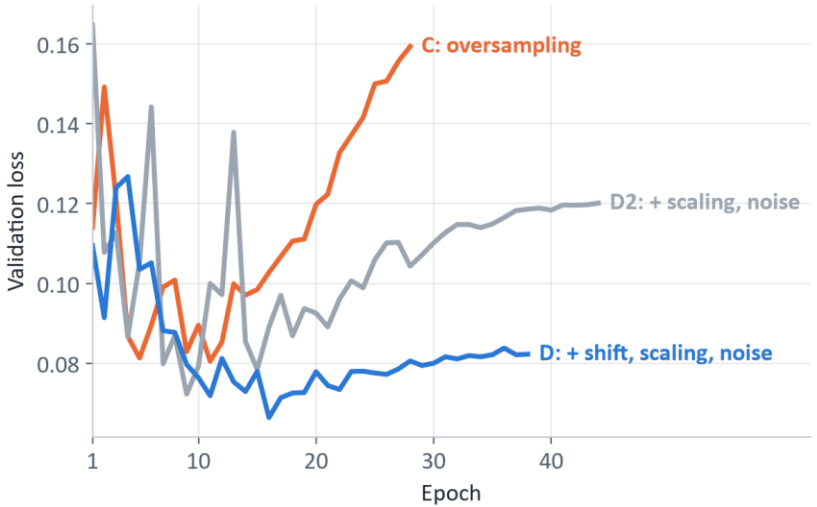
F weight 27: recall of F up, but precision drops to 0.73

C **Oversampling**

Every class copied to 57,977 beats: the copies are memorized

D **+ Augmentation**

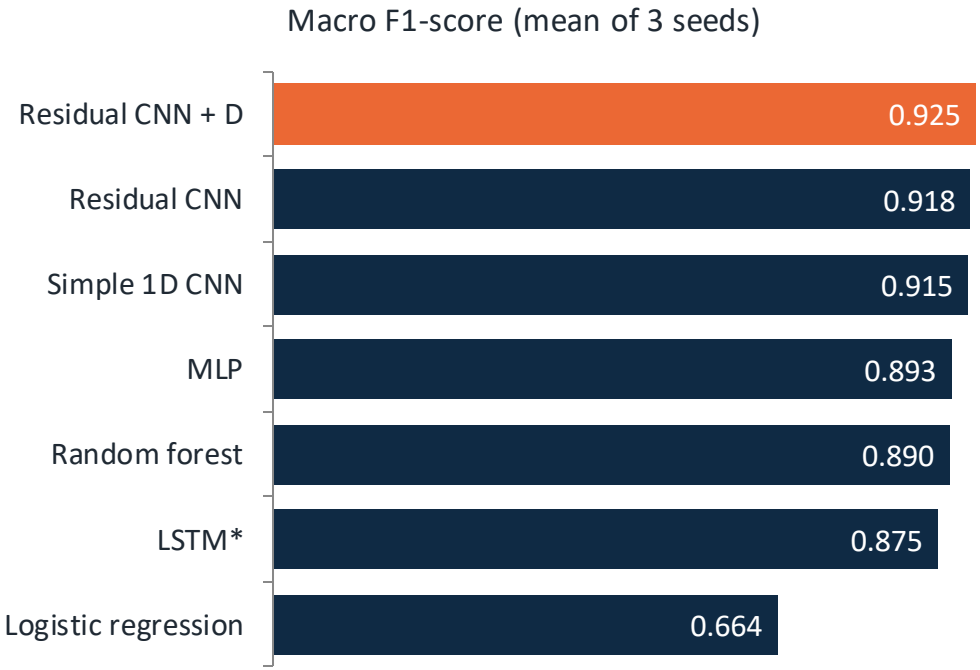
Shift, scaling and noise on every batch: overfitting stops



Validation loss of the residual CNN

Final choice: D, oversampling + augmentation

Results on the test set



0.925

macro F1-score (± 0.004)

93.5%

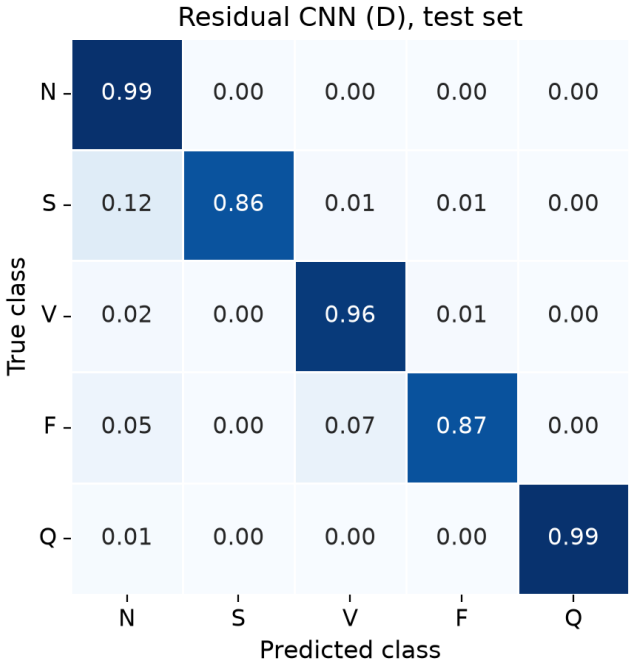
balanced accuracy (± 0.1)

0.86 / 0.87

recall of S / F (was 0.80 / 0.77)

* LSTM: mean of 2 runs; one run never learned

Where does the model make mistakes?



12% of S beats → N

S beats look like normal beats; they differ mainly in timing (they come too early), which a single beat does not show.

7% of F beats → V

A fusion beat is a mixture of a ventricular and a normal beat.

N, V and Q

Recognized almost perfectly (recall of 96% or more).

Comparison and limitations

93.4%

Kachuee et al. (2018)
average accuracy, balanced test set

93.5%

This work
balanced accuracy, official test set

Plain accuracy (98.6%) is high only because 83% of the test beats are normal.

1 **Beat-based split**

Beats of one patient can be in training and test; results are optimistic for new patients.

2 **No rhythm information**

The model sees one beat, not the R-R interval.

3 **LSTM**

Slow on a CPU and unstable: one of three runs failed.

Conclusion

1. CNNs learn the beat shape best: macro F1 0.925, ahead of random forest (0.890) and LSTM (0.875)

2. Oversampling + augmentation mainly helps the rare classes: F recall 0.77 → 0.87, and training is more stable

3. Next steps: inter-patient evaluation and adding R-R interval information

Thank you · Questions? github.com/ahmetertaul/ecg-project